

EE168 - Lab 3 Report

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1 1-bit Full Adder

1.1 Schematic Design

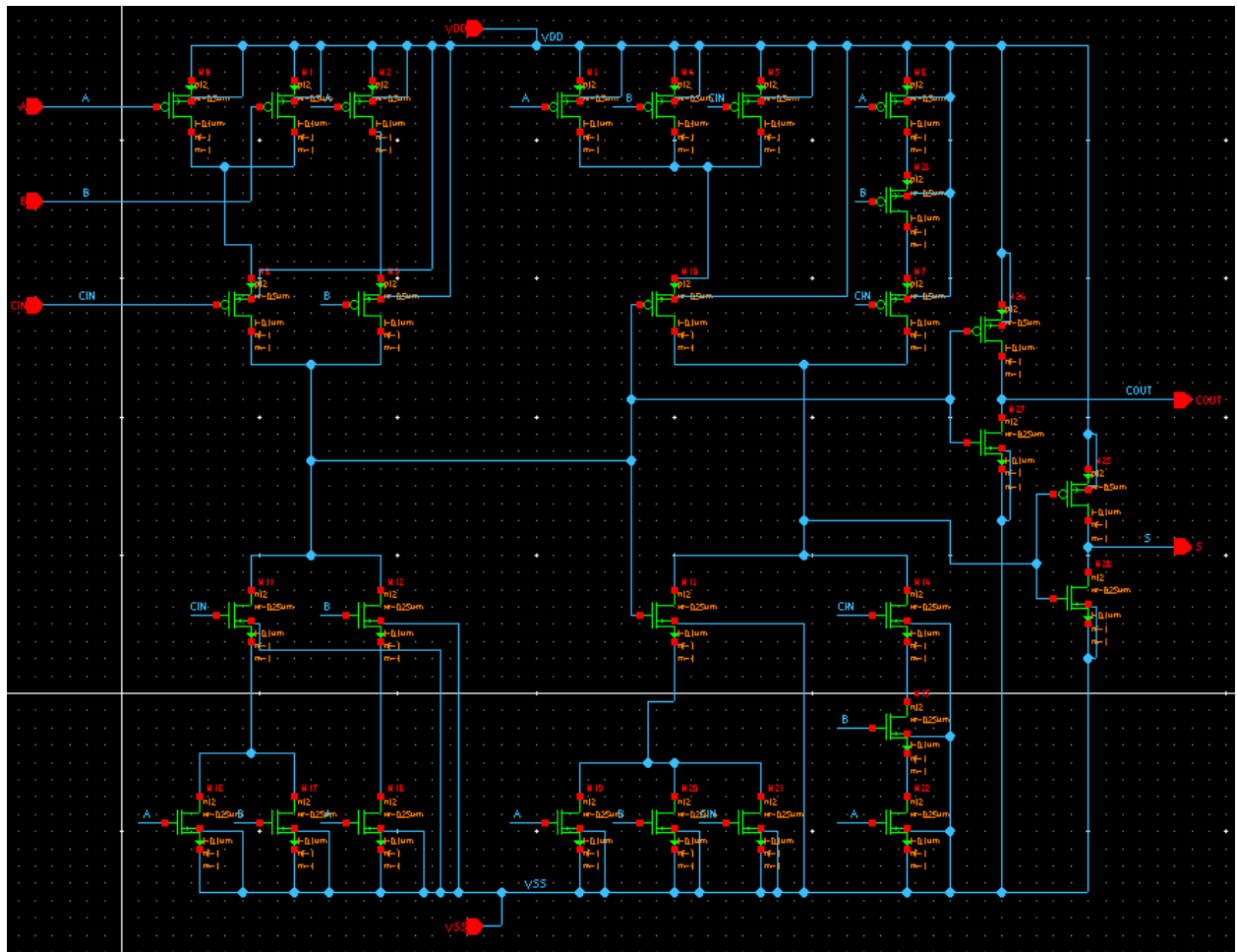


Figure 1: 1-bit Full Adder Schematic

1.2 Layout Design

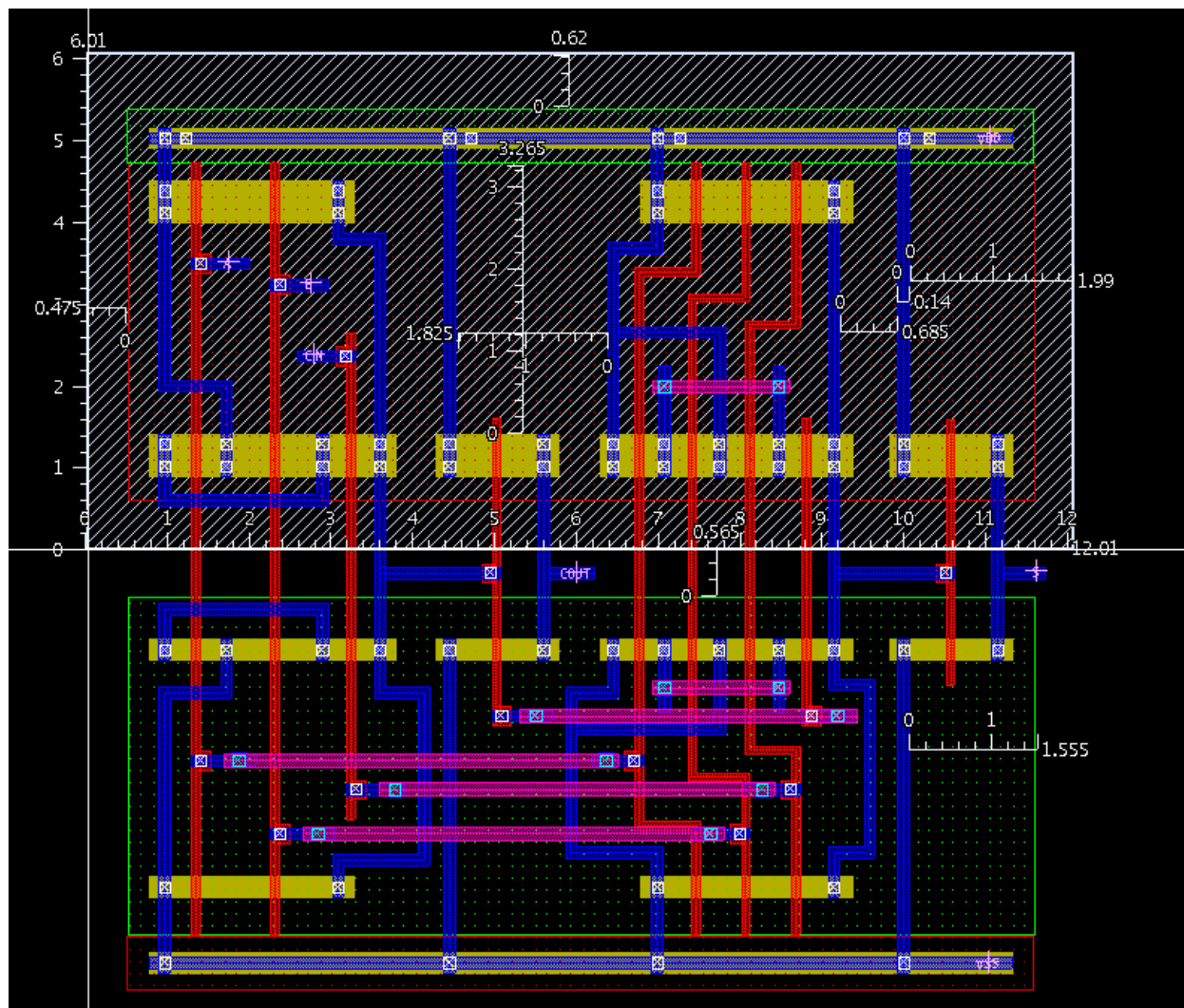


Figure 2: 1-bit Full Adder Layout

1.3 DRC Results

LAYOUT ERRORS RESULTS

CLEAN

DRC Error Statistics

Library name: mylibrary
Structure name: onefulladder
Generated by: IC Validator RHEL64 V-2023.12-SP5-13.12417845 2025/09/16
Runset name: /usr/local/synopsys/pdk/SAED90nm_PDK_10222017/icv/drc/rules.drc.9m_saed90_icv.rs
User name: vvada002
Time started: 2025/12/05 10:35:26PM
Time ended: 2025/12/05 10:35:29PM

Called as: icv -f openaccess -i mylibrary -c onefulladder -oa_view layout -oa_lib_defs /home/eemaj/vvada002/eecs168/

Figure 3: 1-bit Full Adder DRC Results

1.4 LVS Results

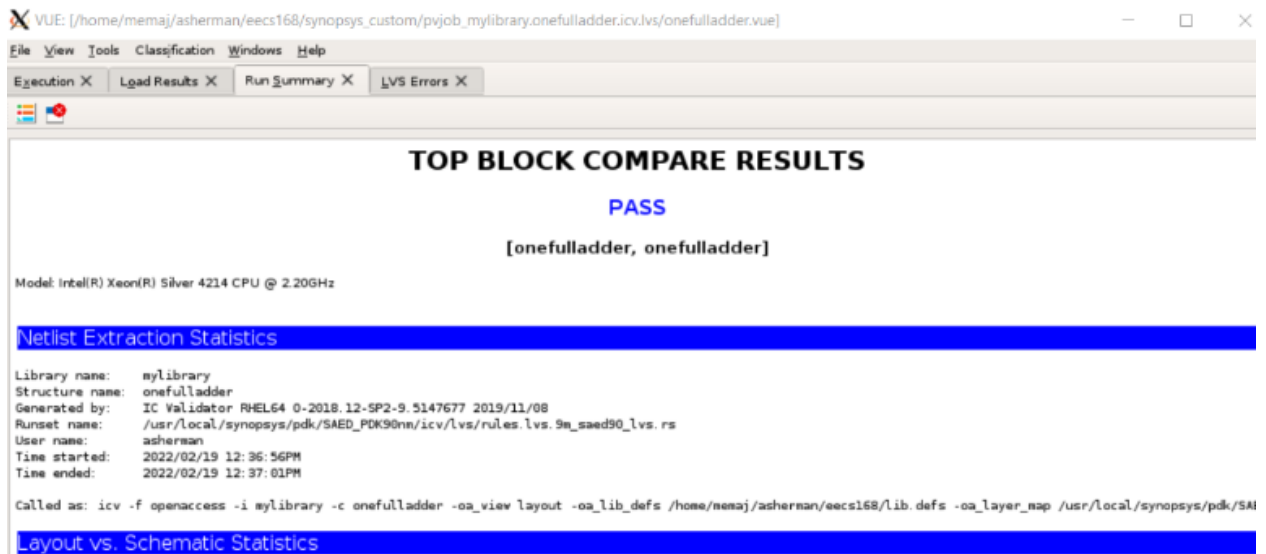


Figure 4: 1-bit Full Adder LVS Results

2 1-bit FA Testbench

2.1 Schematic Design

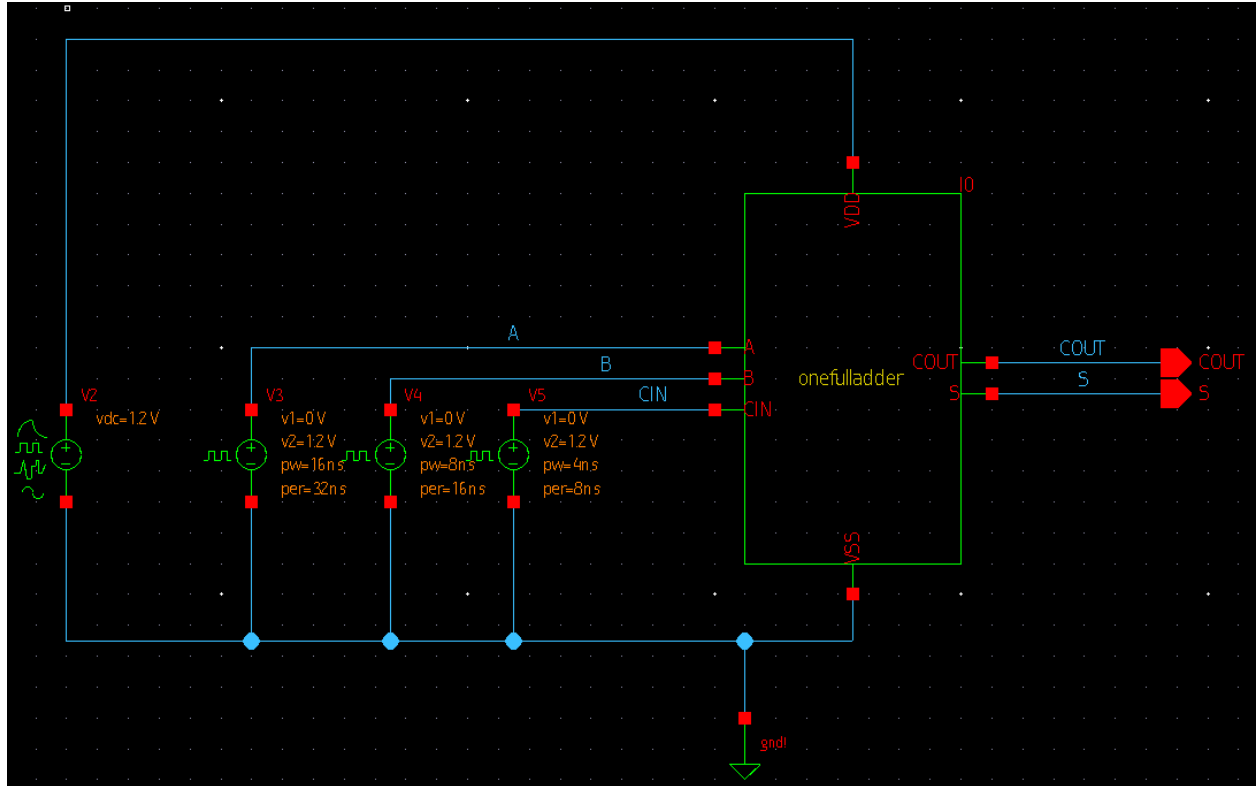


Figure 5: 1-bit Full Adder Testbench Schematic

3 4-bit Full Adder

3.1 Schematic Design

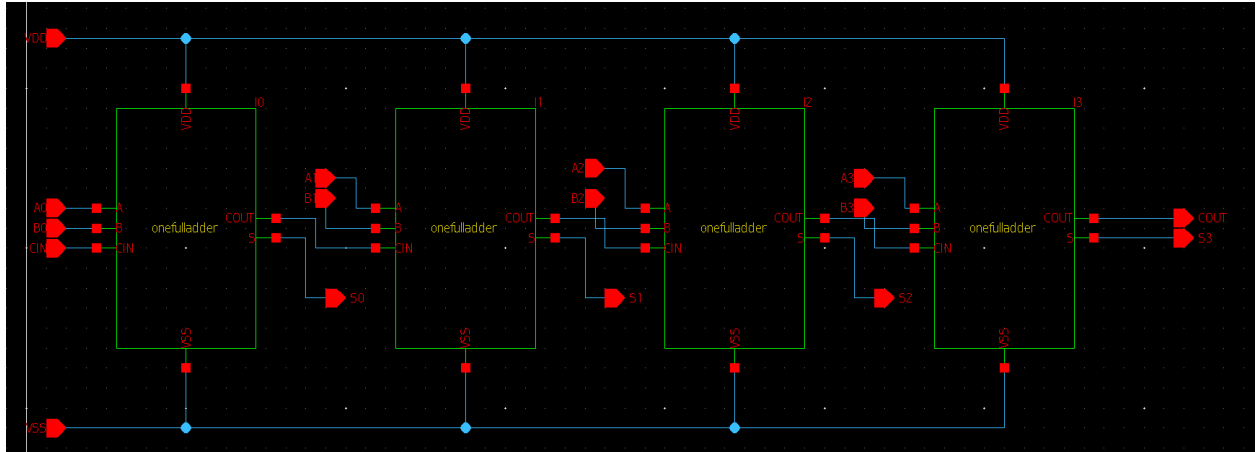


Figure 6: 4-bit Full Adder Schematic

3.2 Layout Design

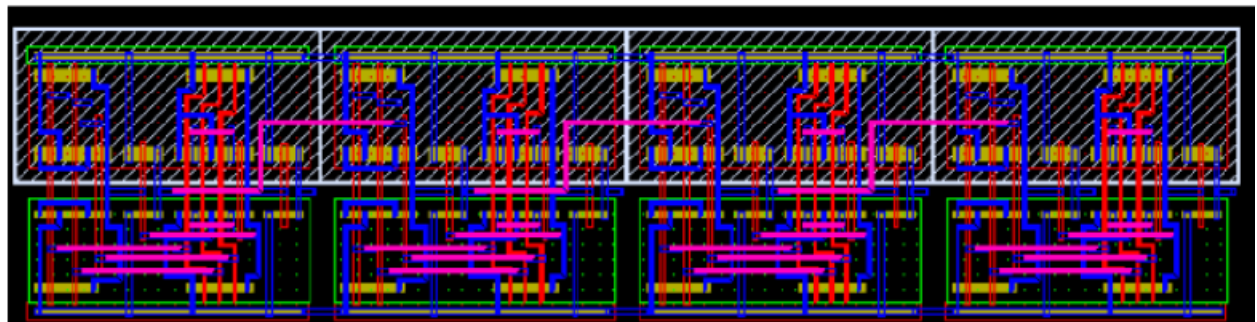


Figure 7: 4-bit Full Adder Layout

3.3 DRC Results

LAYOUT ERRORS RESULTS

CLEAN

DRC Error Statistics

Library name: mylibrary
Structure name: fourfulladder
Generated by: IC Validator RHEL64 V-2023.12-SP5-13.12417845 2025/09/16
Runset name: /usr/local/synopsys/pdk/SAED90nm_PDK_10222017/icv/drc/rules.drc.9m_saed90_icv.rs
User name: vvada002
Time started: 2025/12/05 10:53:13PM
Time ended: 2025/12/05 10:53:17PM

Called as: icv -f openaccess -i mylibrary -c fourfulladder -oa_view layout -oa_lib_defs /home/eemaj/vvada002/eecs16

Figure 8: 4-bit Full Adder DRC Results

3.4 LVS Results

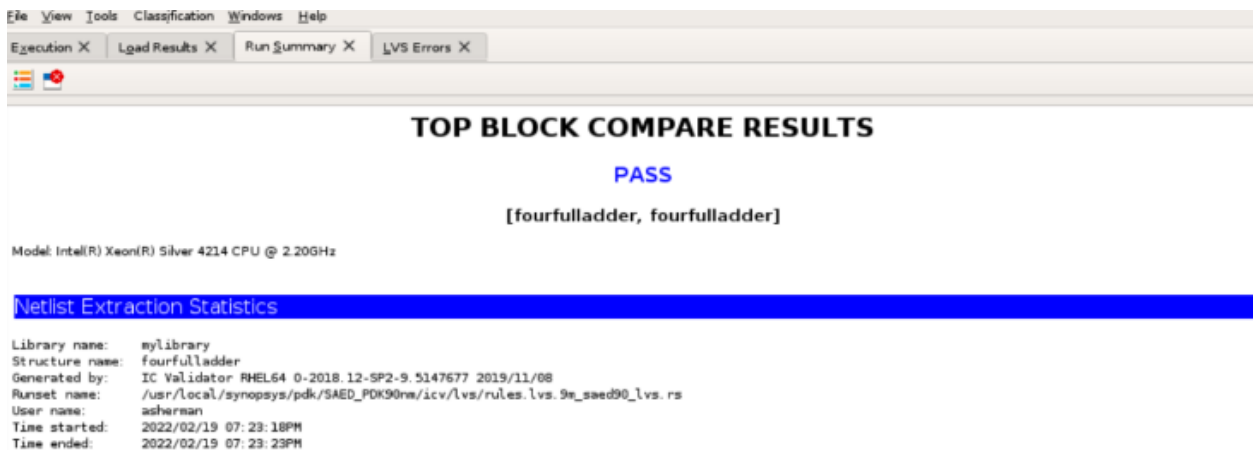


Figure 9: 4-bit Full Adder LVS Results

4 4-bit FA Testbench

4.1 Schematic Design

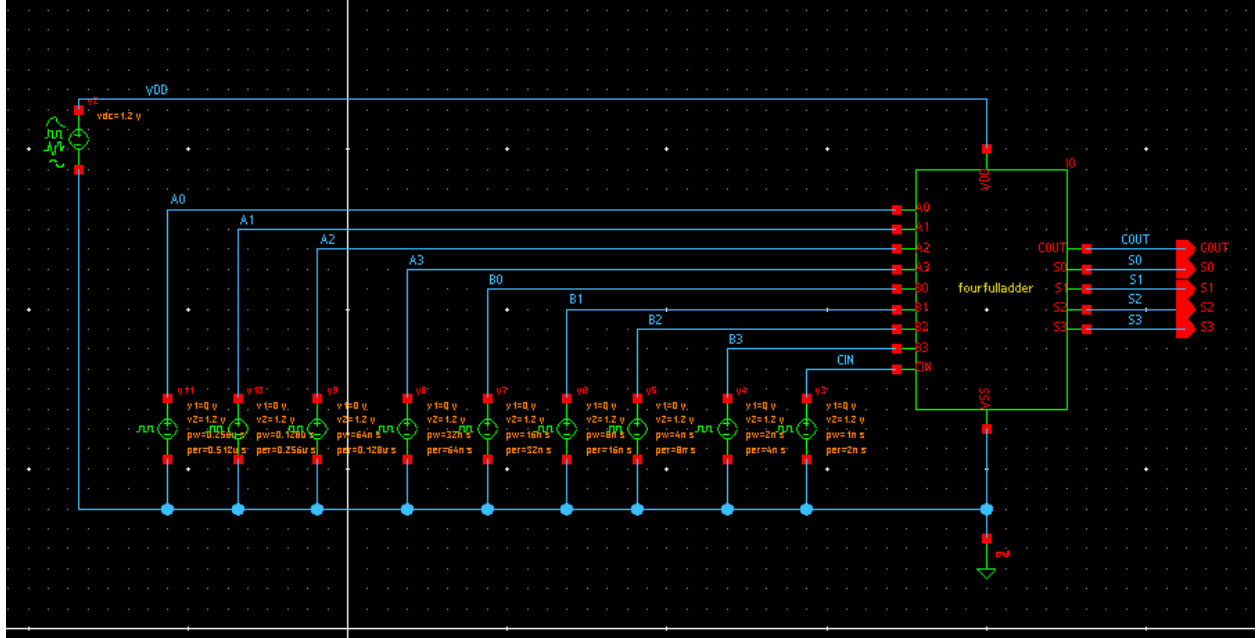


Figure 10: 4-bit Full Adder Testbench Schematic